

A CH Counterexample to the WLD–Dunford–Pettis Projective Tensor Question

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Status

This is a candidate counterexample under CH to Question 8(b) of Aviles, Martinez-Cervantes, Rodriguez, and Rueda Zoca, *Topological properties in tensor products of Banach spaces*, arXiv:2202.00371.

Source Question

Question 8(b) asks:

Suppose that X and Y are WLD and that either X or Y has the Dunford-Pettis property.
Is $X \widehat{\otimes}_\pi Y$ WLD?

Claim

Theorem 1. *Assume CH. Then there exist WLD Banach spaces X and Y , with X having the Dunford-Pettis property, such that*

$$X \widehat{\otimes}_\pi Y$$

is not WLD. In particular, under CH, Question 8(b) has a negative answer.

Idea of the Proof

The source paper already isolates exactly what can go wrong. Its Theorem 4.1 says that $X \widehat{\otimes}_\pi Y$ is WLD if and only if X and Y are WLD and every operator in both cross directions

$$X \rightarrow Y^*, \quad Y \rightarrow X^*$$

has norm-separable range.

Now take a WLD space Y whose dual ball fails property (M). Under CH, such a Y exists by the Plichko/Plebanek construction quoted in the source paper. The source paper's Lebesgue-Bochner corollary says that, for WLD Y , property (M) of (B_{Y^*}, w^*) is equivalent to $L_1(\mu, Y)$ being WLD for every finite measure μ . Hence failure of property (M) supplies a finite measure μ for which $L_1(\mu, Y)$ is not WLD. Since $L_1(\mu, Y) \cong L_1(\mu) \widehat{\otimes}_\pi Y$, and $L_1(\mu)$ is WCG hence WLD and has the Dunford-Pettis property, this is the desired counterexample.

Proof

We use three facts recorded in the source paper.

- (1) The source paper's Theorem 4.1 states that $X \widehat{\otimes}_\pi Y$ is WLD if and only if X and Y are WLD and

$$\mathcal{L}(X, Y^*) = \mathcal{S}(X, Y^*), \quad \mathcal{L}(Y, X^*) = \mathcal{S}(Y, X^*),$$

where $\mathcal{S}$ denotes the operators with norm-separable range.

- (2) The source paper's Theorem 4.4 and Corollary 4.5 say that, for a Banach space Y , the following are equivalent:

$$Y \text{ is WLD and } (B_{Y^*}, w^*) \text{ has property (M)}$$

and

$$L_1(\mu, Y) \text{ is WLD for every finite measure } \mu.$$

- (3) In its preliminaries on measures on Corson compact spaces, the source paper recalls that under CH there exist WLD Banach spaces Y for which (B_{Y^*}, w^*) fails property (M), citing Plebanek's Corollary 4.4.

Assume CH, and choose such a WLD Banach space Y . By (2), since (B_{Y^*}, w^*) fails property (M), it is not true that $L_1(\mu, Y)$ is WLD for every finite measure μ . Therefore there is a finite measure μ such that $L_1(\mu, Y)$ is not WLD.

Set

$$X := L_1(\mu).$$

Since μ is finite, X is weakly compactly generated. Indeed,

$$\{f \in L_1(\mu) : |f| \leq 1\}$$

is weakly compact by uniform integrability, and the linear span of this order interval contains $L_\infty(\mu)$, which is dense in $L_1(\mu)$. Hence X is WCG and therefore WLD. Also, $L_1(\mu)$ has the Dunford-Pettis property by the classical Dunford-Pettis theorem.

Finally, the standard Bochner-space identification gives an isometric isomorphism

$$L_1(\mu, Y) \cong L_1(\mu) \widehat{\otimes}_\pi Y = X \widehat{\otimes}_\pi Y.$$

By the choice of μ , this space is not WLD. Thus X and Y are WLD, one factor X has the Dunford-Pettis property, but $X \widehat{\otimes}_\pi Y$ is not WLD.

Equivalently, applying (1), the same example supplies an operator

$$L_1(\mu) \rightarrow Y^*$$

with nonseparable range. This is precisely the obstruction forbidden by the positive conclusion of Question 8(b).

Relation to the Existing Partial Result

The earlier packet

$$\text{solutions/partial/2202.00371.c0_projective_tensor_wld_subcase/}$$

proves in ZFC that $c_0(\Gamma) \widehat{\otimes}_\pi Y$ is WLD for every WLD Y . The present packet is complementary: it shows that under CH, the arbitrary WLD+Dunford-Pettis factor case has a negative answer. It does not invalidate the $c_0(\Gamma)$ positive subcase.

Novelty and Literature Check

The run’s cheap indexes contained the previous $c_0(\Gamma)$ partial packet for arXiv:2202.00371 but no packet for this $L_1(\mu)/\text{CH}$ counterexample.

Local source searches and web searches were run for the exact source question and close phrases including:

“Question 8(b)” + “Topological properties in tensor products of Banach spaces”,

“WLD” + “Dunford-Pettis” + “projective tensor product” + “property (M)”,

and

“Lebesgue-Bochner” + “WLD” + “property (M)”.

They found the source arXiv page and adjacent tensor-product/Dunford-Pettis literature, but no later direct formulation of this CH counterexample.

This packet should therefore be treated as a short CH consequence of the source paper’s own characterizations together with the CH example quoted from Plebanek, rather than as an already-explicit literature answer.

Limitations

This is a counterexample under CH, not a ZFC counterexample. The ZFC status of Question 8(b) remains separate. In models where all Corson compact spaces have property (M), this particular construction is unavailable.

References

- [1] A. Aviles, G. Martinez-Cervantes, J. Rodriguez, A. Rueda Zoca, *Topological properties in tensor products of Banach spaces*, arXiv:2202.00371.
- [2] G. Plebanek, *Convex Corson compacta and Radon measures*, Fund. Math. **175** (2002), no. 2, 143–154.
- [3] J. Diestel, *A survey of results related to the Dunford-Pettis property*, Contemporary Mathematics **2** (1980), 15–60.

Human Review Recommendation

Review as a likely valid counterexample under CH. The most important audit points are:

- (i) the source Corollary 4.5 quantifier “for every finite measure μ ”;
- (ii) the CH existence of a WLD Y whose dual ball fails property (M), as quoted in the source preliminaries;
- (iii) the standard facts that finite-measure $L_1(\mu)$ spaces are WCG/WLD and have the Dunford-Pettis property.